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ME-1847

B. Tech (First Semester)
EXAMINATION, 2025-26

**COMPUTER CONCEPT AND
PROGRAMMING IN 'C'**

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed. All Sections (A, B and C) are mandatory

Section—A

(Very Short Answer Type Questions)

Note : Attempt any seven questions. (Each question carries 8 marks. $7 \times 8 = 56$)

1. Define Algorithm. Write all the characteristics of an Algorithm.

P. T. O.

[2]

Second ~~Third~~ symbol
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~~1.~~ Convert the following numbers :

(i) $(127.35)_8 = (?)_{10}$

(ii) $(12345.48)_{10} = (?)_{16}$

~~2.~~ Explain the role of operating system.

1.5 x 2

~~3.~~ Write notes on bitwise logical operators.

7.3.0

~~4.~~ Write notes on printf() and scanf() functions.

~~5.~~ Write note on storage classes.

$5 \times 0.5 + 1.5 \times 2 + 2 + 5$
 $2.5 + 3.0 + 2.5$

~~6.~~ Evaluate the following expression and show their hierarchy :

$\Rightarrow 10 + 12 = 22$

$A = 5. * 1/2 + 3/2. * 2 + 2 + 5.5;$

(Assume A to be an int)

0.5×5
 2.5 Morning

~~7.~~ Explain the following character input and output file function :

(i) getc()

(ii) putc()

$4 \overline{) 30} 7$
 28
 20
 20

~~8.~~ Write a program in C to read a file and display the contents of the file on the screen using fgetc() function.

$5/2 + 30/4 + 2 + 5.5;$

- ~~10.~~ WAP in C that swaps values of two variables.

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 12 marks. $2 \times 12 = 24$

- ~~1.~~ With the help of an example explain the dangling if-else problem.
- ~~2.~~ Write a program in C using switch case to display a menu that offers five options :
read three numbers, calculate total, calculate average, display the smallest and display the largest value.
- ~~3.~~ Write a C program which demonstrates the use of break, and continue statements.
- ~~4.~~ WAP in C to print the sum of all odd numbers from 1 to 50.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. Write a program in C, program accepts two matrices of the same order and finds the sum of these matrices and prints the sum matrix.

2. Develop a program in C to create a postal address system :

Name

Address

City

State

Pin Code

struct

3. Write a C program to find the desired element in an array of N elements. Use pointer for searching the element.

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4. Explain the following :

(i) calloc () function

(ii) malloc () function

(iii) realloc () function

(iv) free () function

x x x x x

Dynamic
Memory
Locations